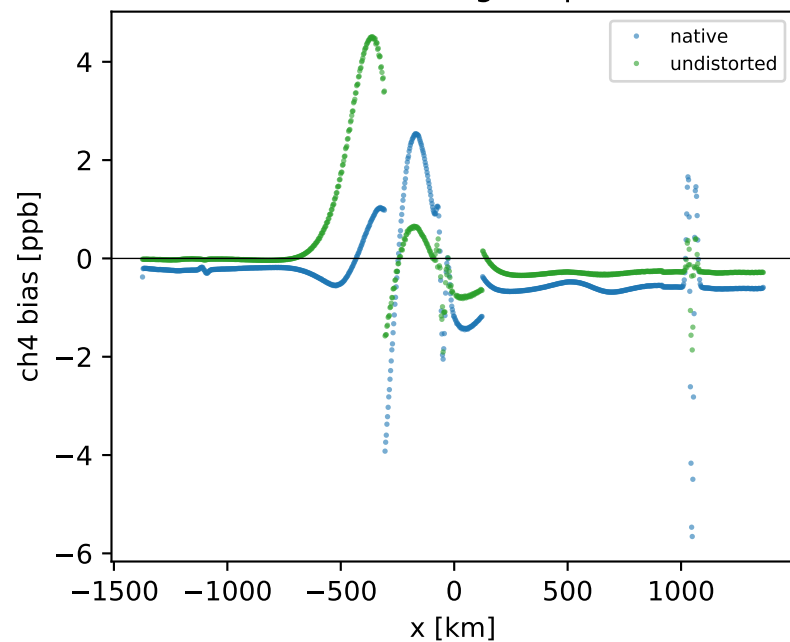
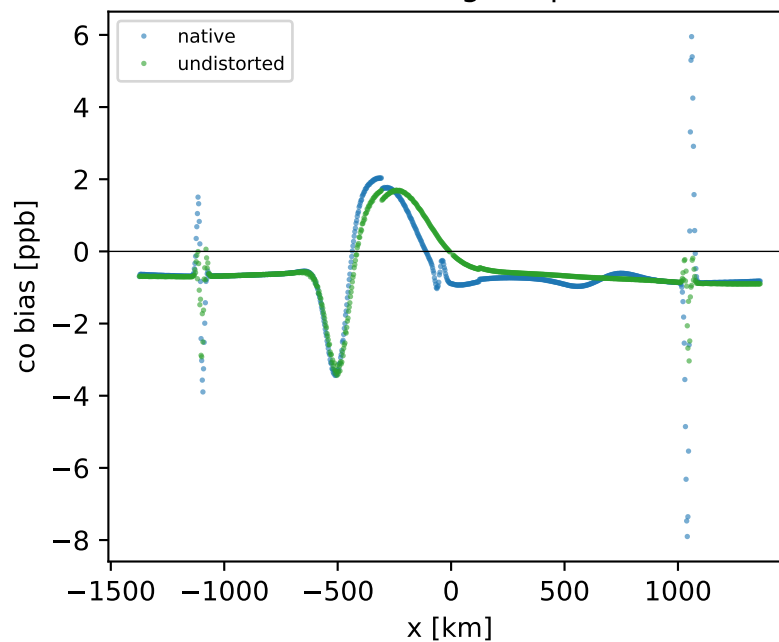


Joint FPA0+FPA3 -- realistic, no noise (native/undistorted)

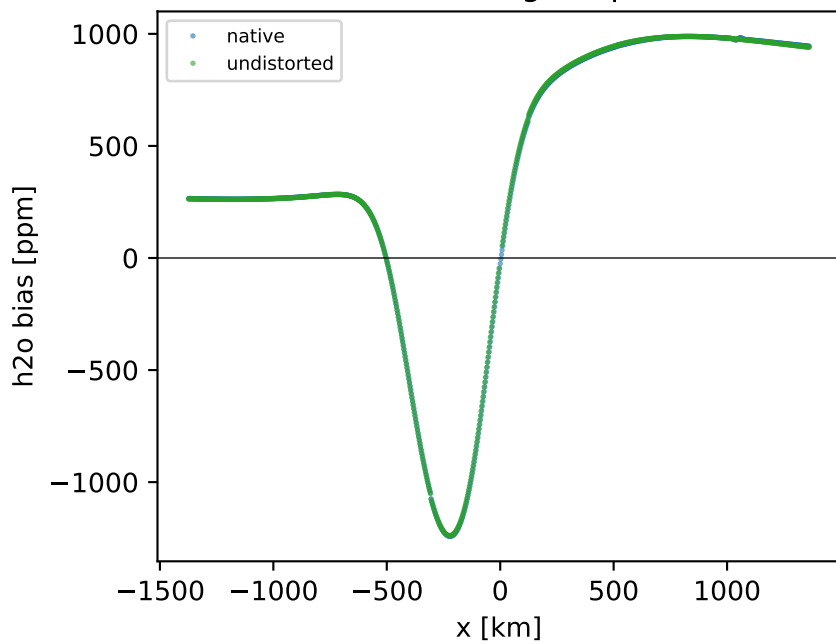
ch4 bias vs along-slit position



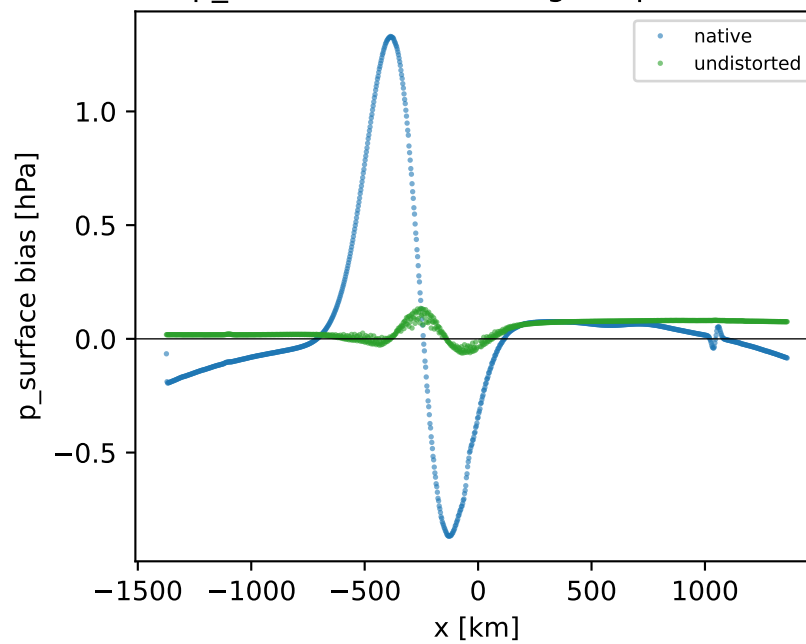
co bias vs along-slit position



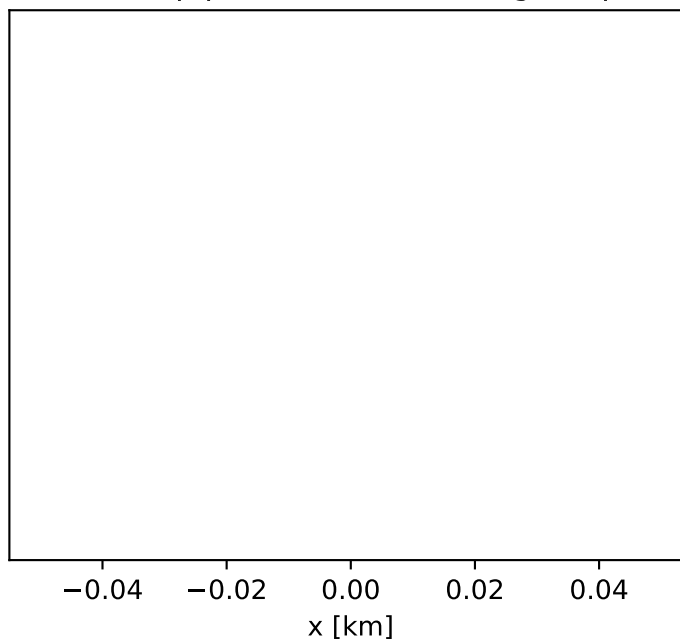
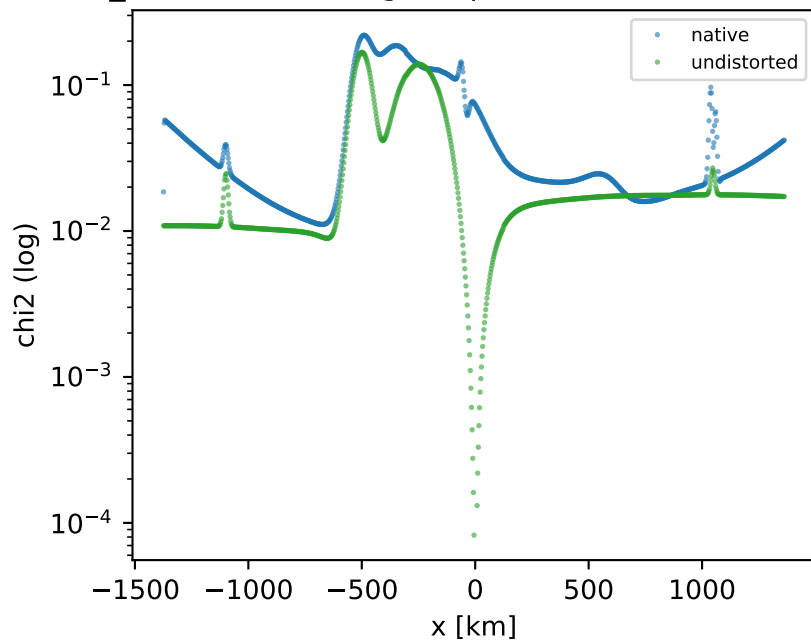
h2o bias vs along-slit position



p\_surface bias vs along-slit position



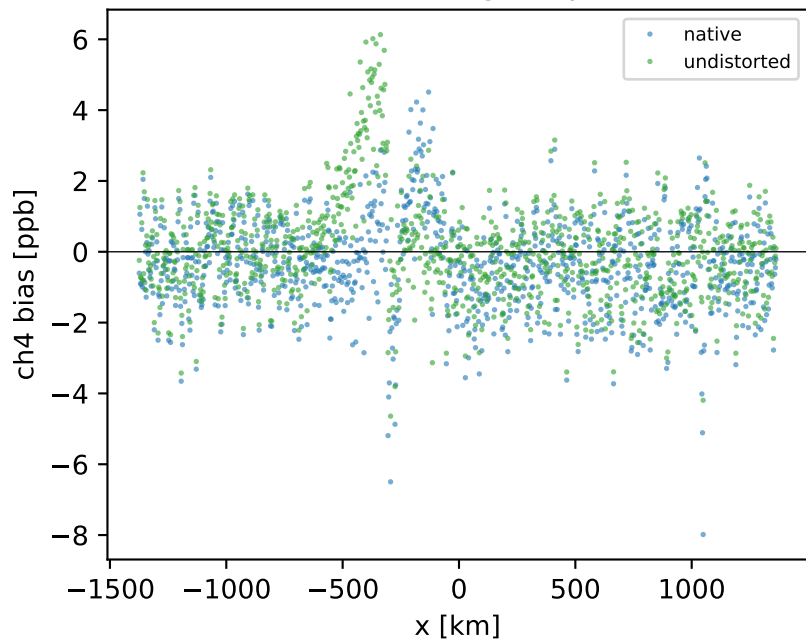
chi2\_reduced vs along-slit position (outliers excluded) Where each pipeline fails, vs along-slit position



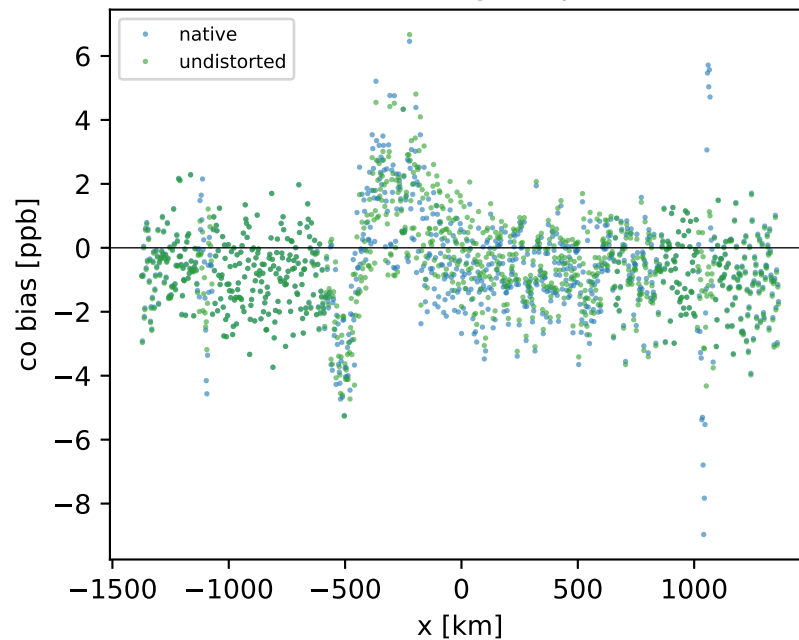
undistorted chi2-outlier (n=4)

# Joint FPA0+FPA3 -- realistic, noise (native/undistorted)

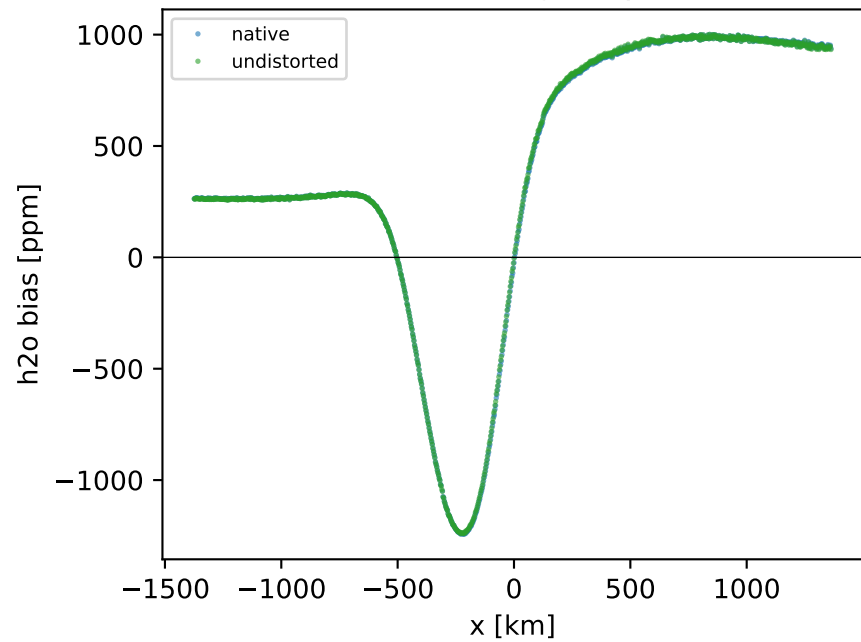
## ch4 bias vs along-slit position



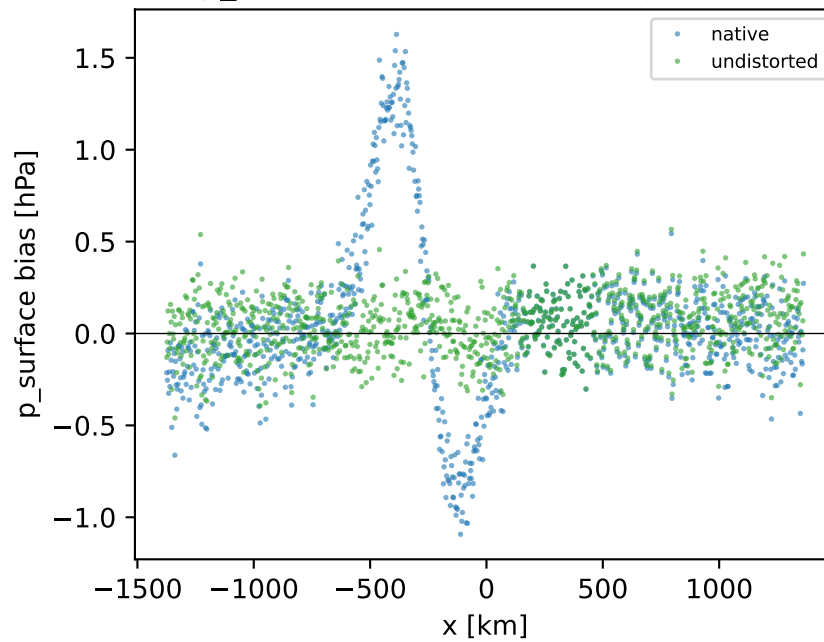
## co bias vs along-slit position



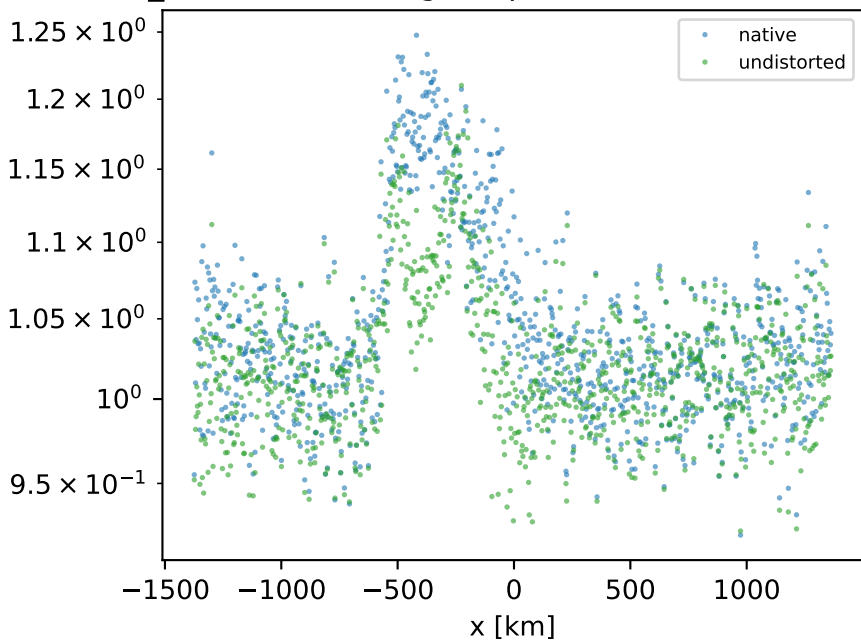
## h2o bias vs along-slit position



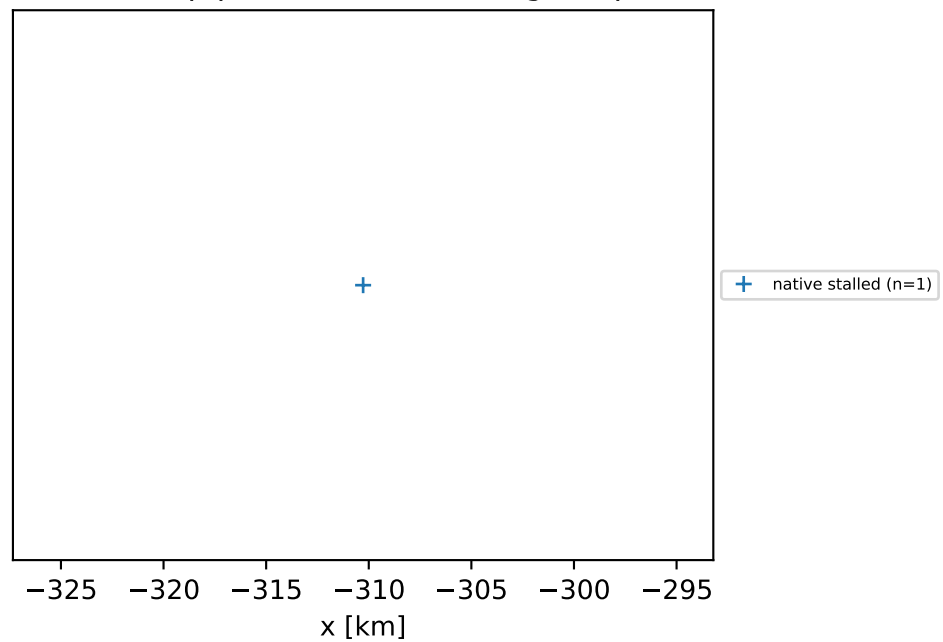
## p\_surface bias vs along-slit position



## chi2\_reduced vs along-slit position (outliers excluded)

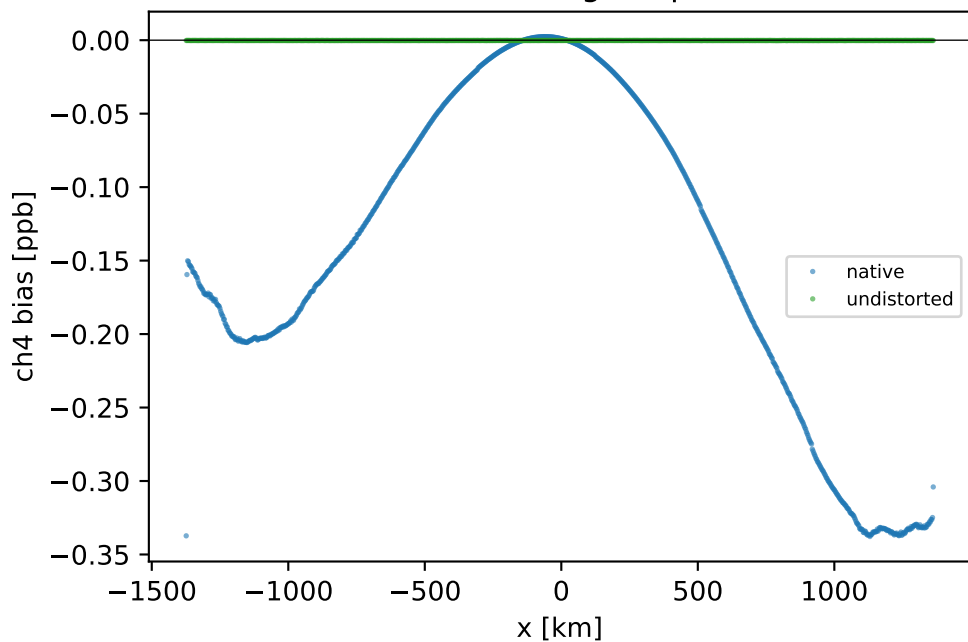


## Where each pipeline fails, vs along-slit position

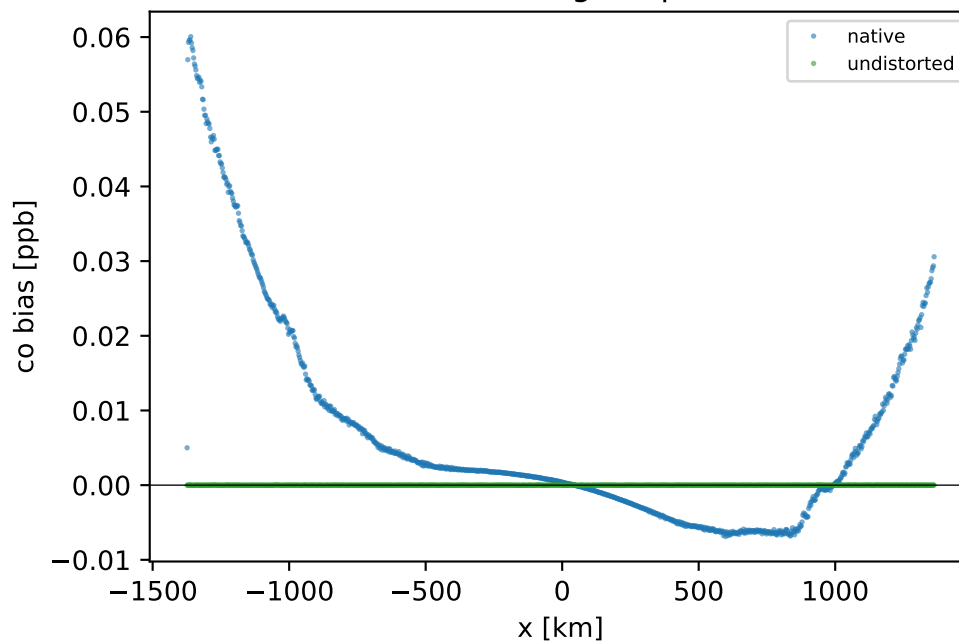


# Joint FPA0+FPA3 -- uniform, no noise (native/undistorted)

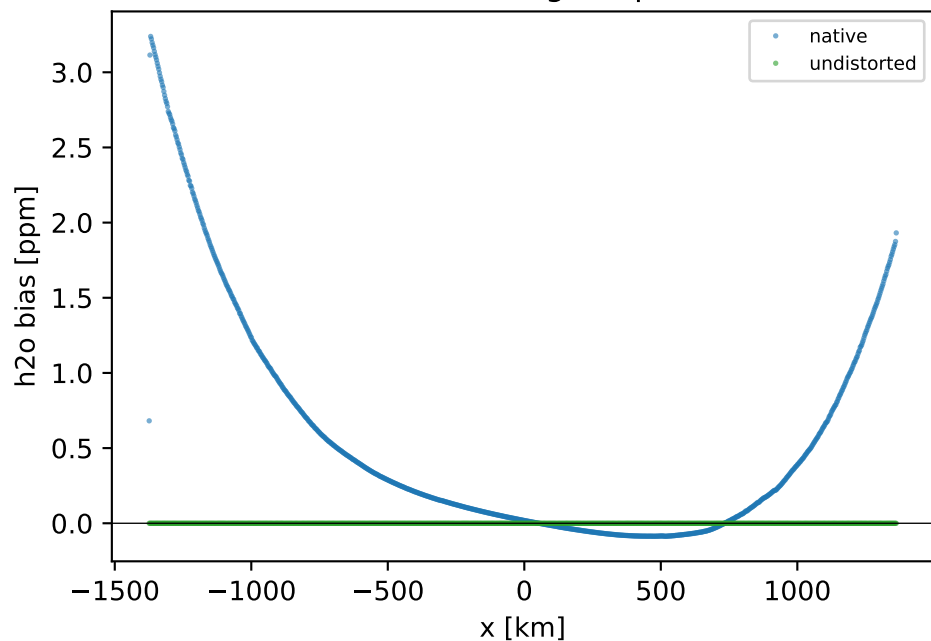
## ch4 bias vs along-slit position



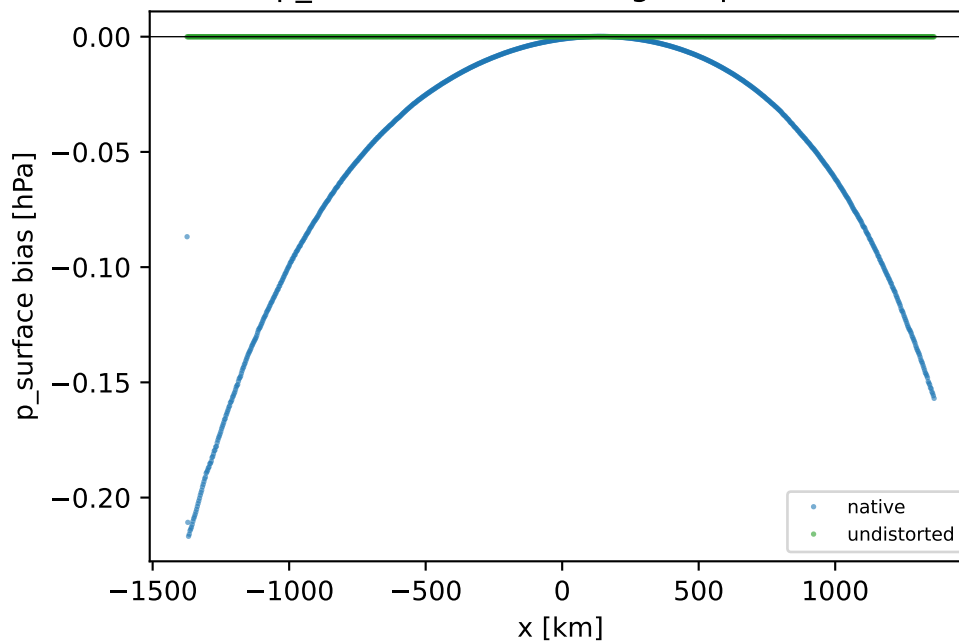
## co bias vs along-slit position



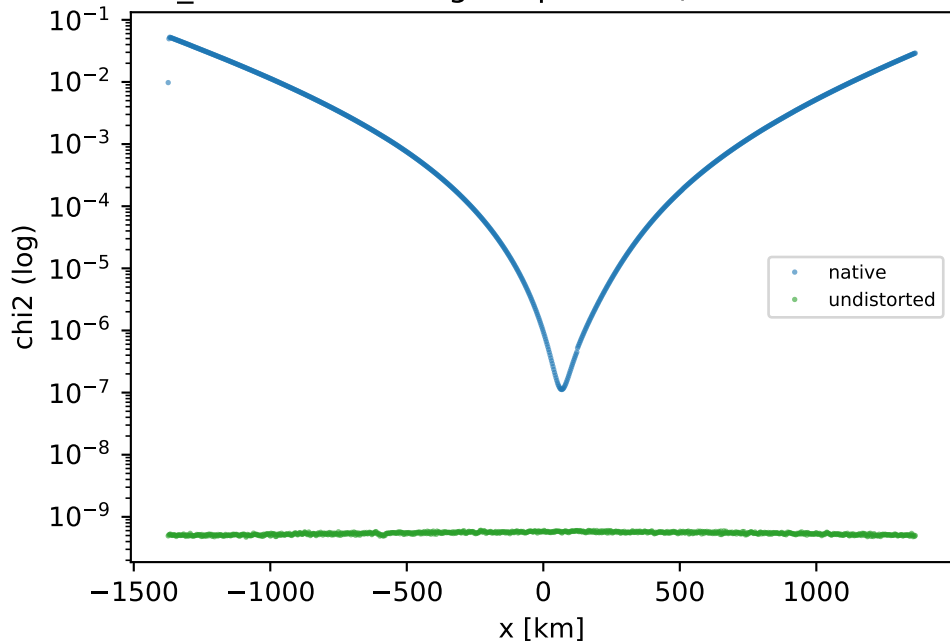
## h2o bias vs along-slit position



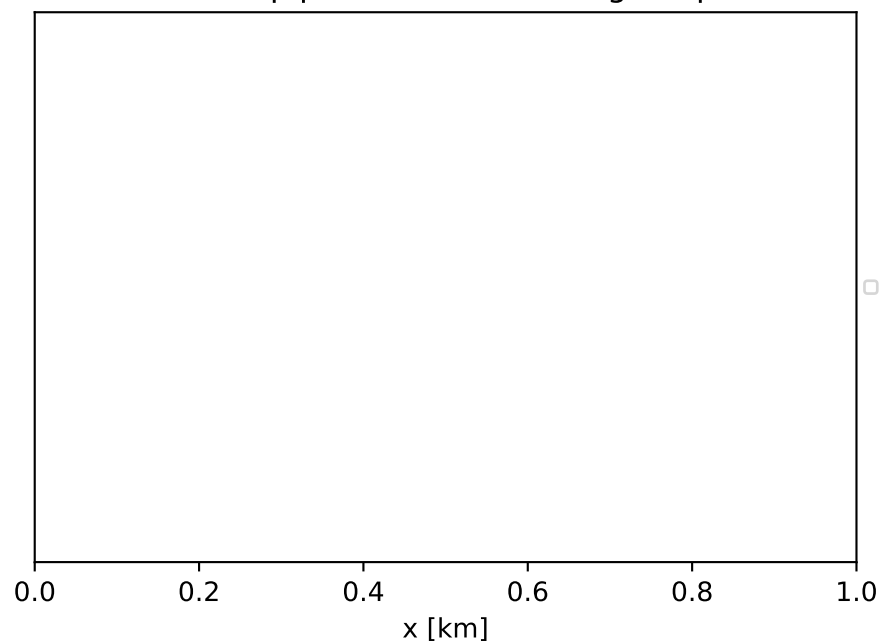
## p\_surface bias vs along-slit position



## chi2\_reduced vs along-slit position (outliers excluded)

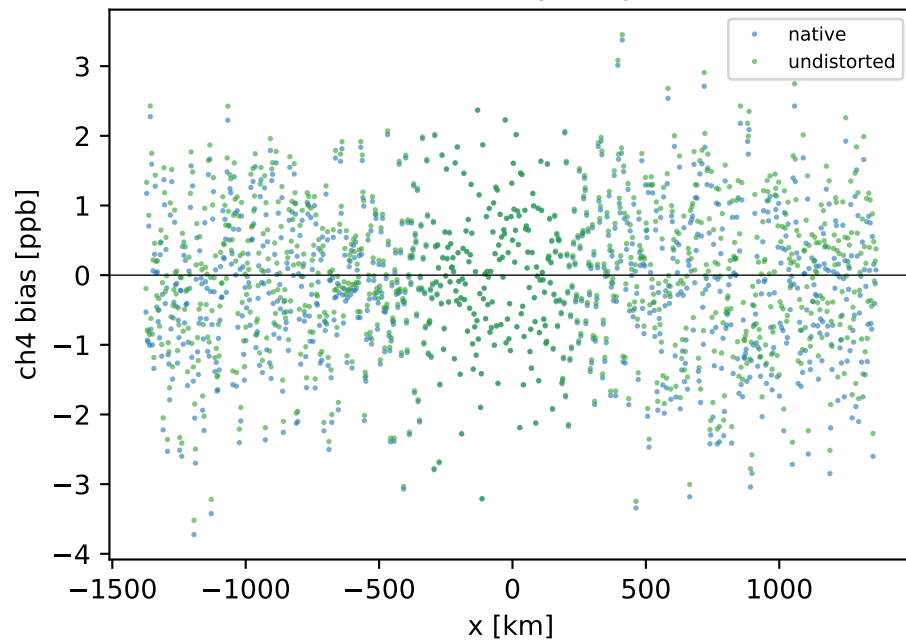


## Where each pipeline fails, vs along-slit position

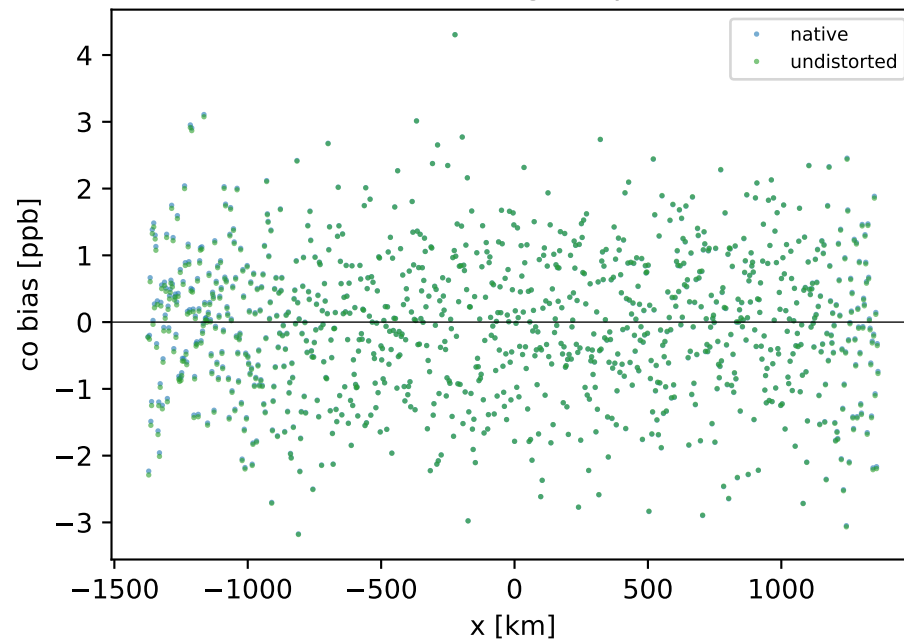


Joint FPA0+FPA3 -- uniform, noise (native/undistorted)

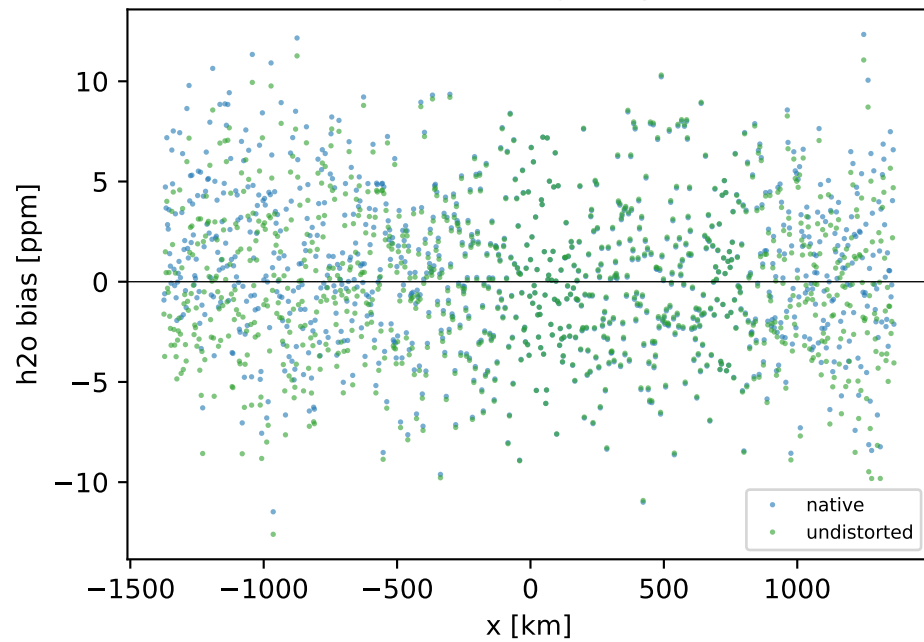
ch4 bias vs along-slit position



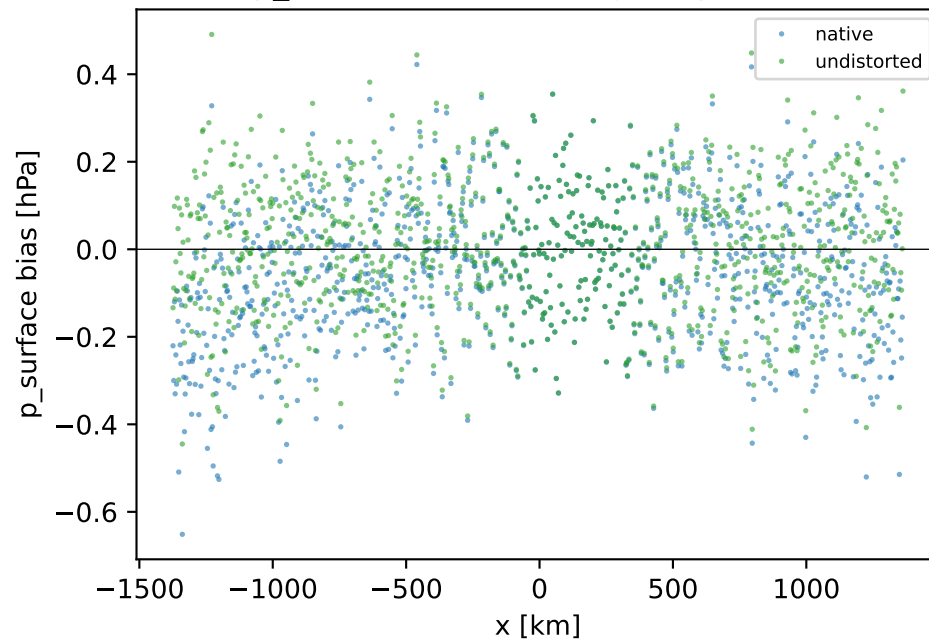
co bias vs along-slit position



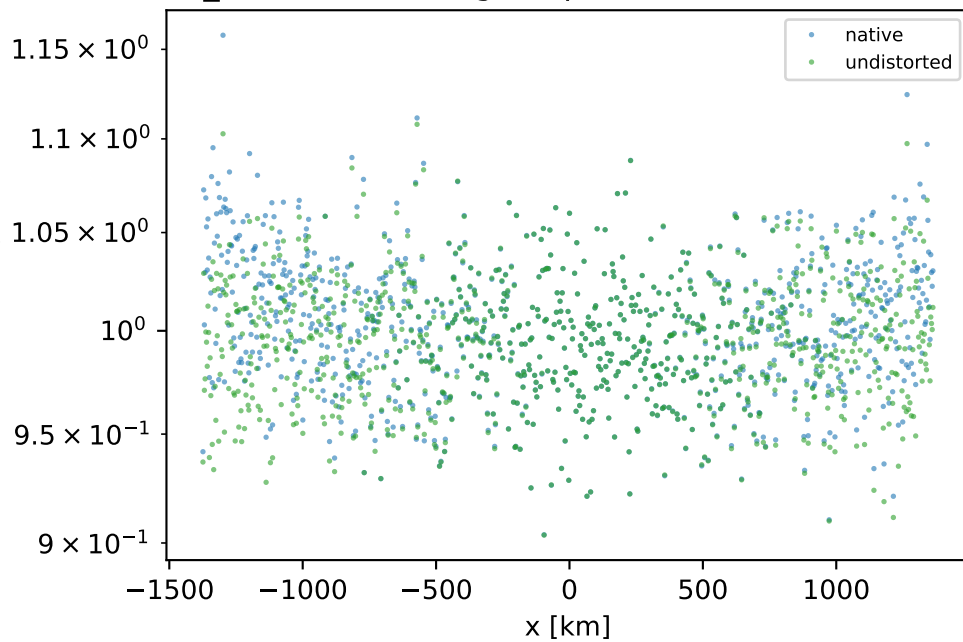
h2o bias vs along-slit position



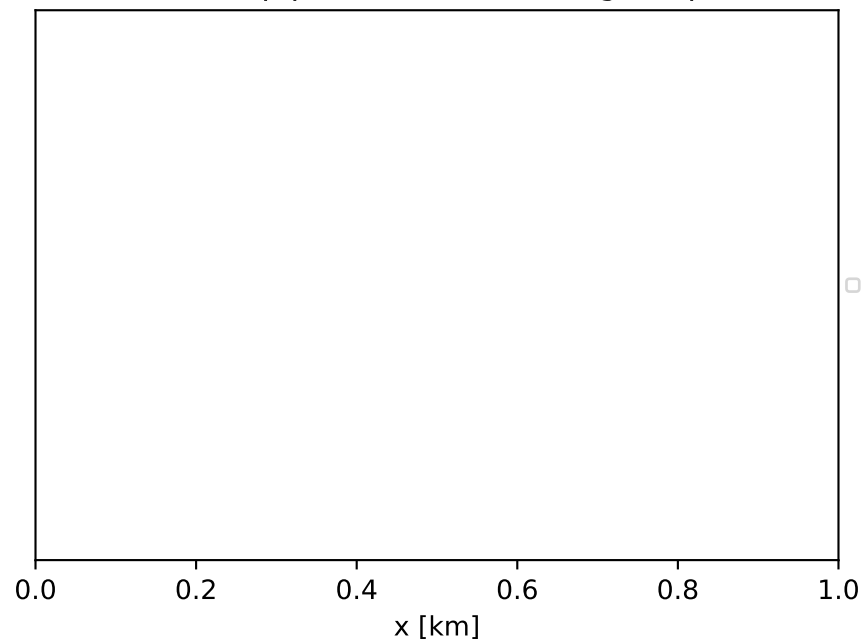
p\_surface bias vs along-slit position



chi2\_reduced vs along-slit position (outliers excluded)

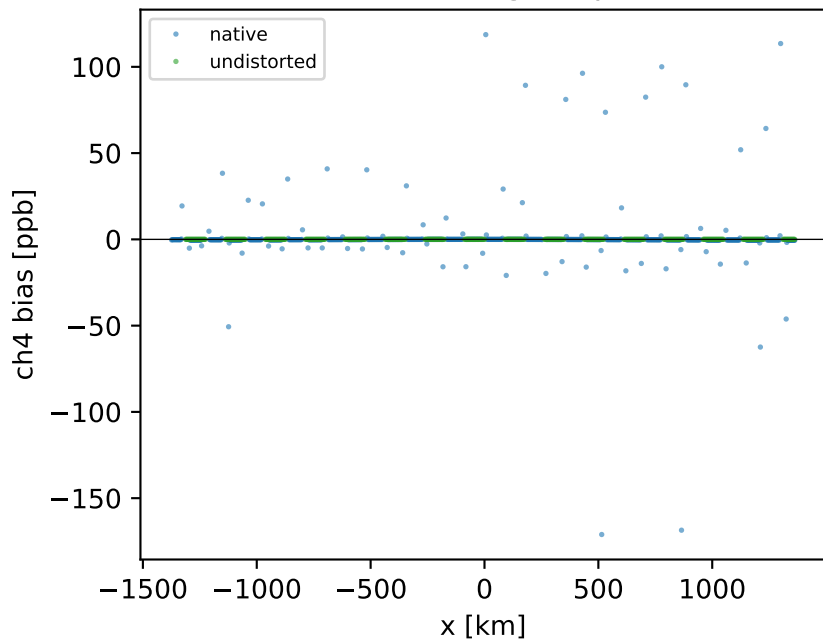


Where each pipeline fails, vs along-slit position

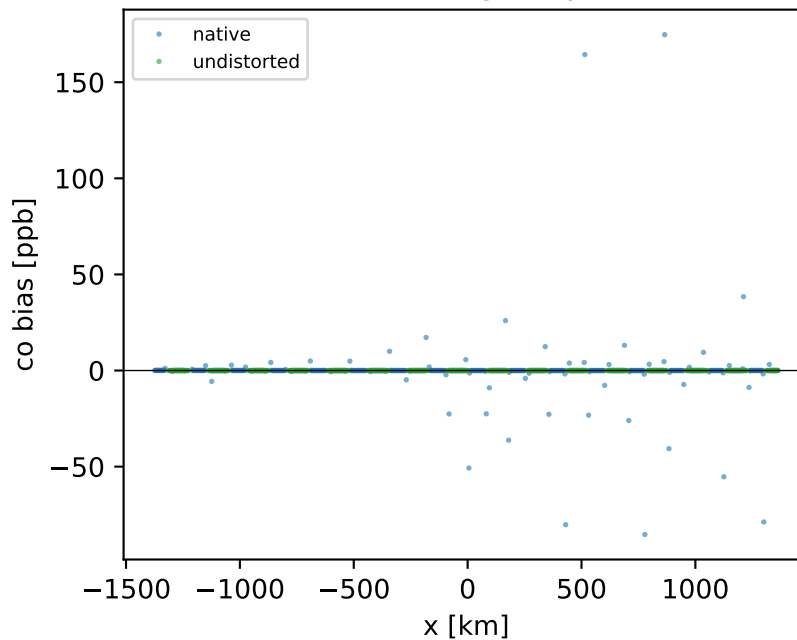


# Joint FPA0+FPA3 -- barcode, no noise (native/undistorted)

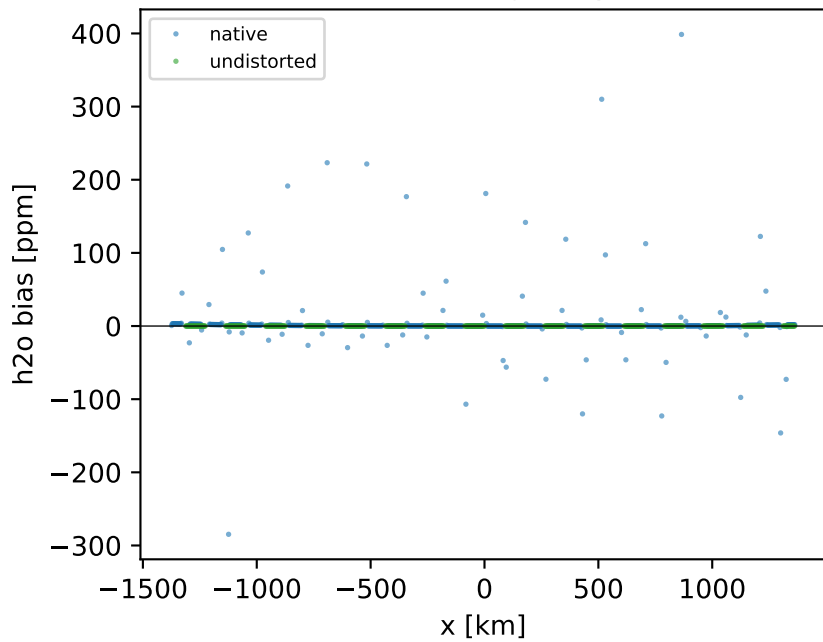
## ch4 bias vs along-slit position



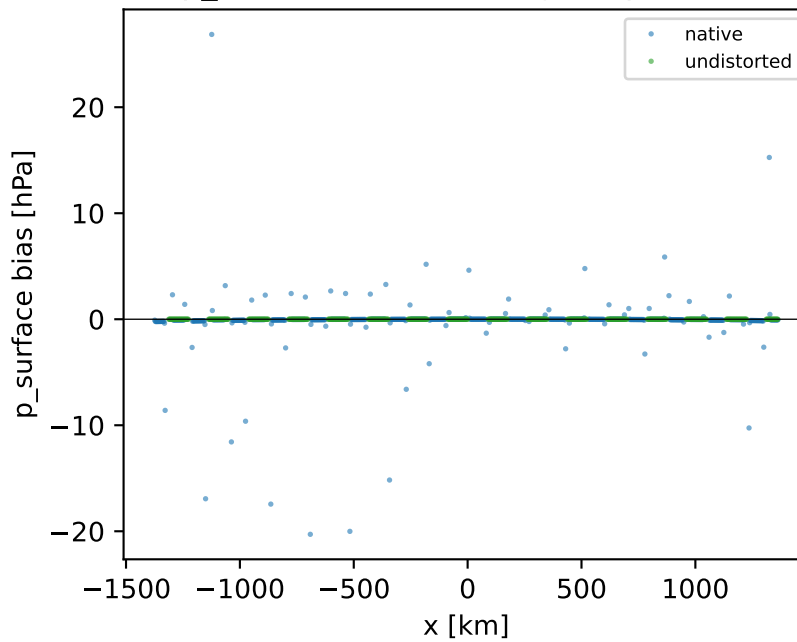
## co bias vs along-slit position



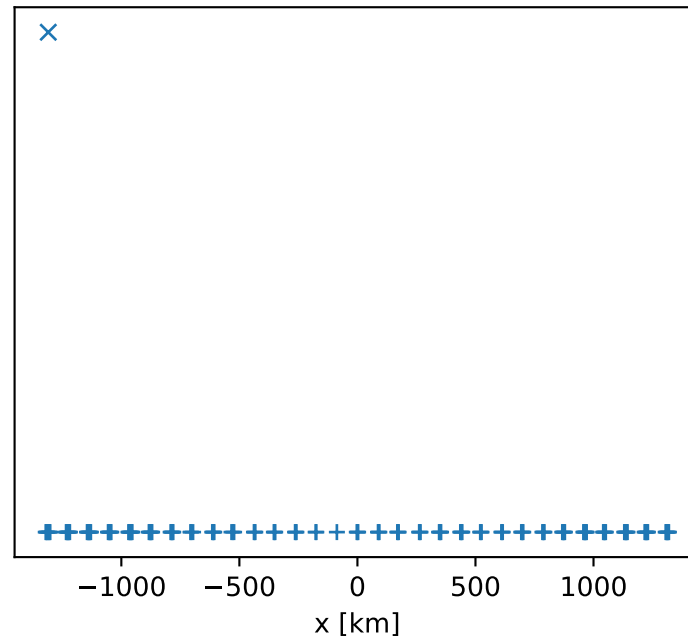
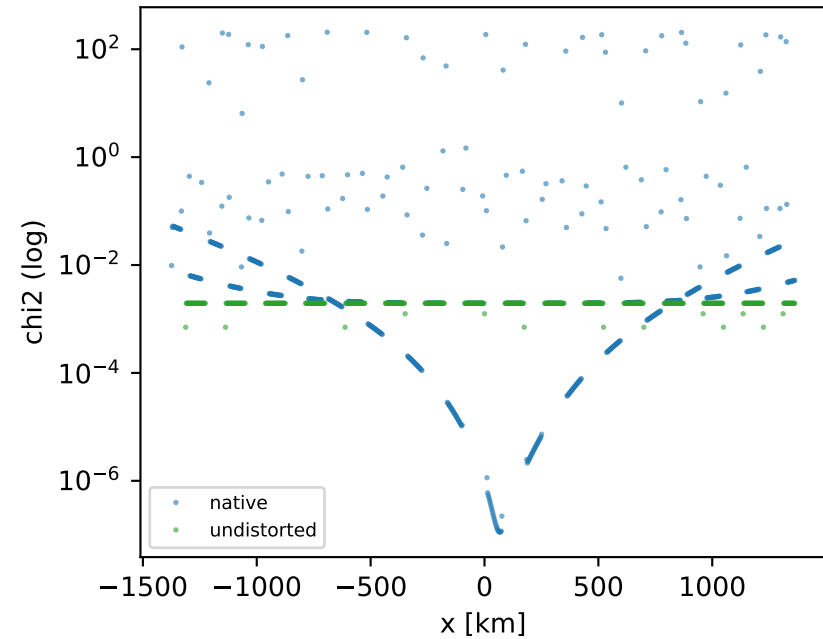
## h2o bias vs along-slit position



## p\_surface bias vs along-slit position

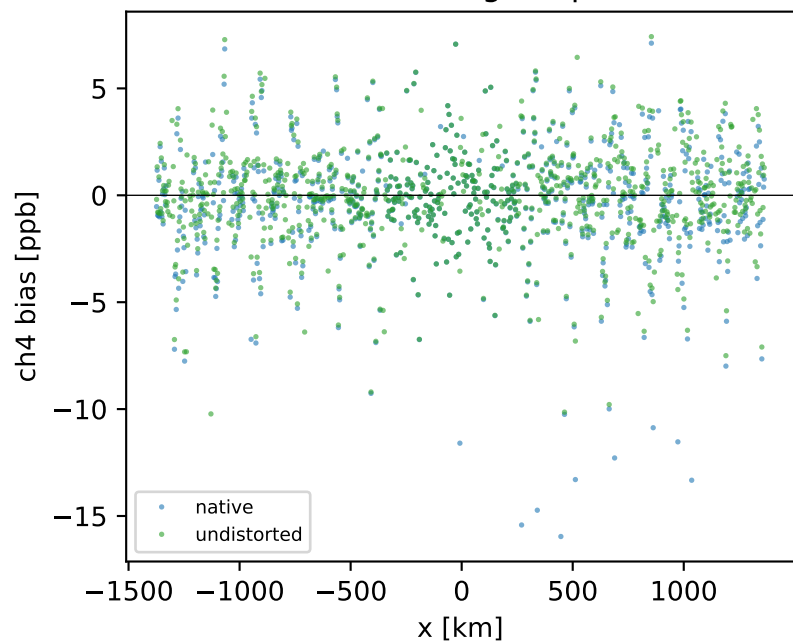


## chi2\_reduced vs along-slit position (outliers excluded) Where each pipeline fails, vs along-slit position

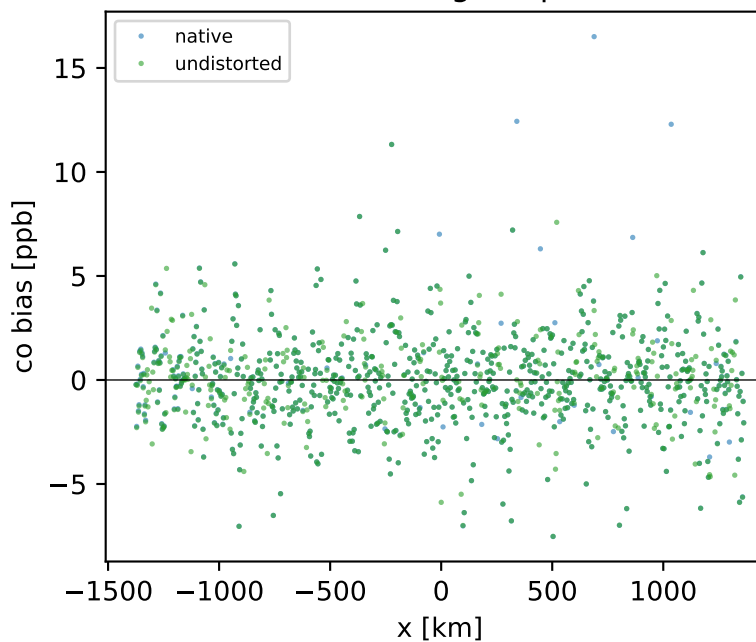


# Joint FPA0+FPA3 -- barcode, noise (native/undistorted)

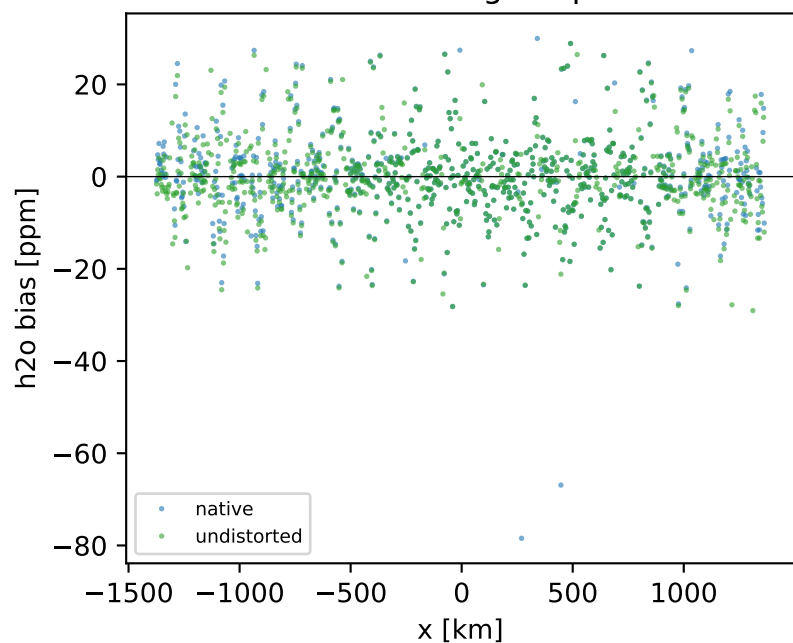
## ch4 bias vs along-slit position



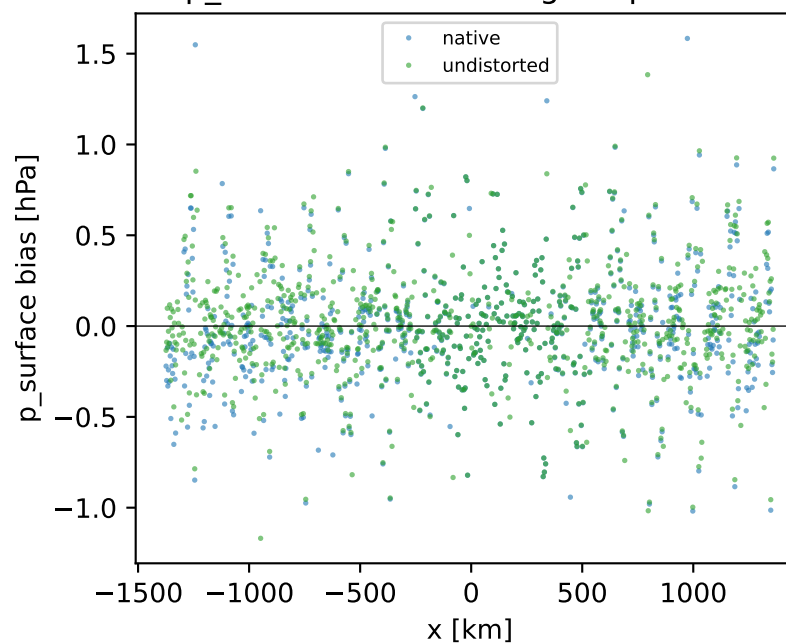
## co bias vs along-slit position



## h2o bias vs along-slit position



## p\_surface bias vs along-slit position



## chi2\_reduced vs along-slit position (outliers excluded) Where each pipeline fails, vs along-slit position

